

Phase 3: Activity Summary

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1. Activity Overview

In Phase 3, we successfully improved our model's diagnostic performance and interpretability by developing and integrating novel, clinically-relevant temporal fluency features. This initiative followed our Phase 2 success, where we achieved a first-place ranking on the competition leaderboard.

Our approach was motivated by the finding that our classification model was already learning to identify influential audio regions. We engineered three specific fluency features, all of which showed a statistically significant difference between healthy controls and individuals with ADRD, confirming their diagnostic value. These features were then used to provide additional supervision to our model through an enhanced architecture, yielding tangible gains in cross-dataset validation.

Our work provides strong evidence that a focus on detailed temporal speech patterns is a key avenue for future performance gains. Furthermore, our bias analysis underscores the critical importance of creating demographically balanced datasets and employing standardized data collection methods to mitigate batch effects and ensure robust, generalizable models.

2. Data

We successfully developed and validated a novel speech fluency scoring model that shows a significant correlation with cognitive decline. By fine-tuning a Qwen2.5-omni-3b model on the expert-annotated Mispeech/speechocean762 dataset, we created a tool that accurately predicts fluency scores.

When applied to the DementiaBank dataset, as shown in Figure 1, the model's predicted scores revealed a statistically significant downward trend from Healthy Controls to individuals with MCI and ADRD (one-way ANOVA, $p < .001$). This confirms the model's ability to capture clinically meaningful changes in speech.

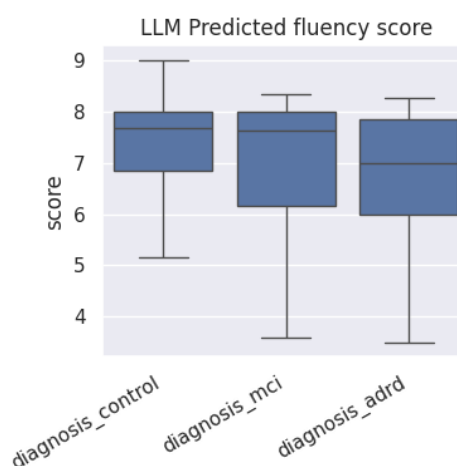


Figure 1, Fluency score predicted by Qwen2.5-omni

This work was part of a broader benchmarking effort using a leave-one-corpus-out policy on DementiaBank's five largest corpora (WLS, PITT, IVANOVA, VAS, and DELAWARE) to ensure the robustness of our findings on limited and varied data.

3. Generalizability

Auxiliary Learning Improves Generalizability and Interpretability.

The multi-task architecture incorporating an auxiliary head to predict temporal features (such as word confidence or fluency) proved to be a highly effective strategy. This approach not only consistently outperformed the baseline classifier across most datasets but also served as a form of regularization. More significantly, it enhanced model interpretability by generating "fluency maps" that visualize specific speech events influencing predictions, thereby directly addressing the "black box" problem prevalent in medical AI applications.

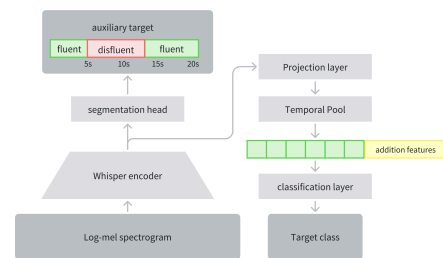


Figure2, the model architecture with auxiliary target

As illustrated in Figure 2, the updated model structure features an additional segmentation head positioned above the transformer's final hidden state, with fluency serving as the prediction target. The experimental results are presented in Table 1. We employed a leave-one-out cross-validation method to evaluate the model's generalizability across different corpora. With the exception of the Spanish corpus IVANOVA, the log-loss decreased across all datasets, indicating that the auxiliary head significantly improves the model's cross-corpus generalizability.

	WLS (ENG, n=825)	PITT (ENG, n=549)	VAS (ENG, n=100)	delaware (ENG, n=81)	IVANOVA(ESP, n=360)
(N/A) Phase II model	0.859	1.049	1.10	2.09	1.04
Word Conf	0.898	0.880	0.967	2.05	1.07
Fluency Score	0.839	0.869	0.956	1.87	1.11

Furthermore, the auxiliary head enables us to predict temporal fluency, allowing users to examine specific timestamps and identify words where the model detected disfluencies, as well as understand how these decisions were made. The figure below demonstrates how we can compare the model's Class Activation Maps (CAM) with novel fluency feature to gain insight into the model's prediction process and determine whether to accept its predictions. Figure 3 depicts the comprehensive information available to users for decision-making. Figure 3a shows basic audio information, while Figure 3b displays the Class Activation Map, highlighting regions important to the model's decision. Figures 3c and 3d present features completely independent of the model that can serve as external validation tools for checking model predictions. In this case, the two areas highlighted by the model's CAM were also captured by our fluency features: at 6-7 seconds, there is both a low fluency score (Figure 3c) and a detected block (Figure 3d); at 18-22 seconds, another long block is present (Figure 3d). Manual verification confirmed expression issues in each flagged zone, as shown in Figure 3e.

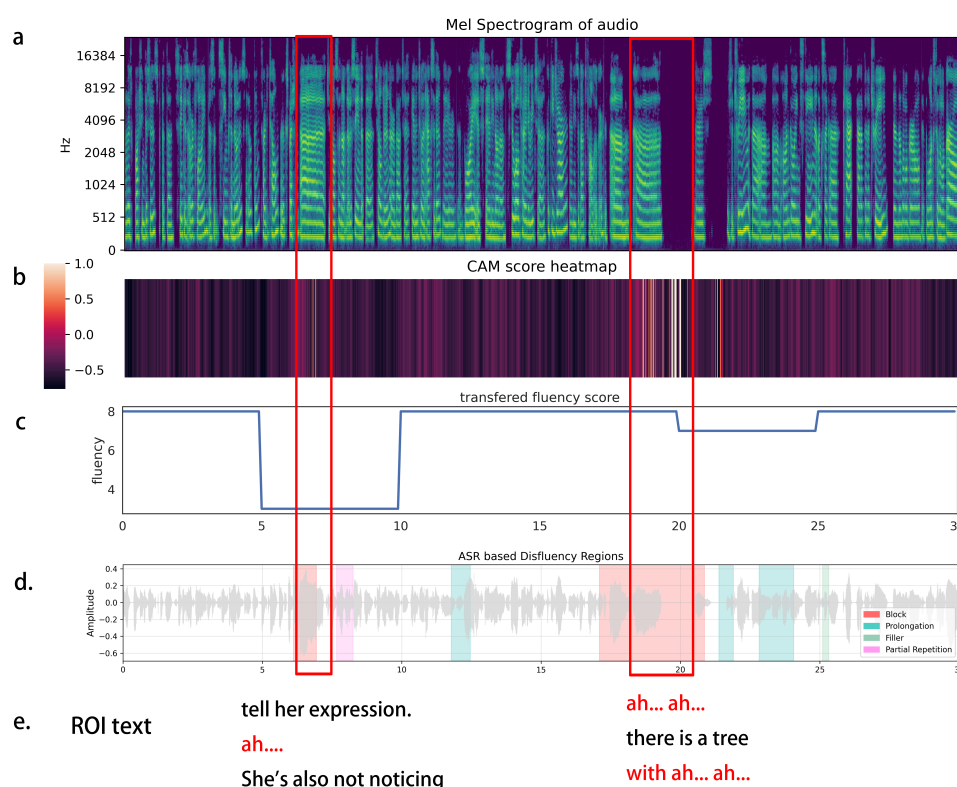


Figure 3, the visualization of the model Interpretability. **a.** the Mel Spectrogram. **b.** The Class activation map, highlight the region where model's decision based on. **c.** the transferred fluency score, which is irrelevant with classification model. **d.** the ASR based disfluency feature, which is also irrelevant with classification model.

4. Methodology

Multimodal LLMs Improve Model Performance

In Phase II, we found that model performance is strongly related to both the number of parameters and the size of the pretraining dataset, with particular dependence on pretraining dataset size. Given the recent prosperity of the LLM community and the availability of numerous multimodal LLMs, we evaluated the state-of-the-art audio model Qwen2.5-omni on our dataset. Using group labels as targets, we fine-tuned the model on the competition dataset and observed improved balanced accuracy from 0.582 to 0.665. This indicates that powerful multimodal LLMs with larger pretraining datasets can significantly enhance performance.

Voice Conversion Shows Potential for Improving Generalizability

We tested a voice conversion tool (SeedVC) to align audio samples from different corpora to a unified voice. However, the results showed a slight decrease in performance (from 0.66 to 0.63 on the competition dataset). We believe this may be limited by the current conversion model's capabilities. We also suggest that this method could be applied to online data collection scenarios.

ASR-based Models Show Potential for Capturing Language-related Features

By training a BERT model, we discovered that in a small subset of cases where speech was fluent but semantically nonsensical, the BERT model predicted a high probability of disease. While this did not improve overall scores, we consider this an interesting finding that warrants further investigation.